

A Benchmark population

60

scenarios

30

families

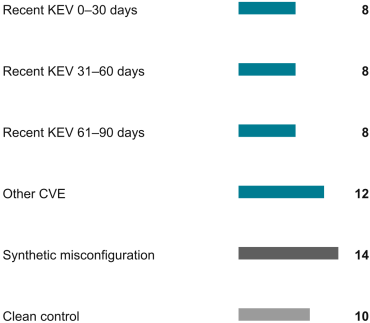
5

repetitions

3,000

cells

Scenario composition

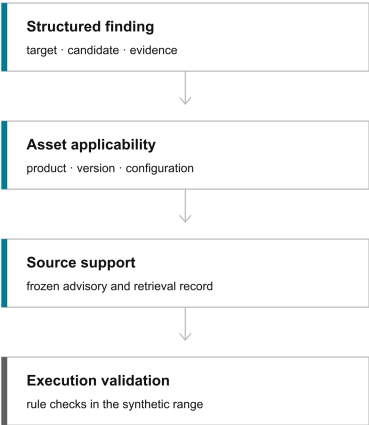


25 positive · 25 paired negative · 10 clean

B Model configurations

Backbone	Single-RAG	Fixed team	Autonomous	Ablation
Qwen 3.5 27B 6 configurations	●	●	●	3
Gemma 3 27B 2 configurations	●	●	—	—
gpt-oss 20B 2 configurations	●	●	—	—
Common controls				
Temperature				0
Seed				paired
Concurrency				1
Runtime egress				0 bytes
Model commands				disabled

C Evidence-verified scoring



TP · FP · FN failure-inclusive

Evidence-verified detection F1

2TP paired family-level inference
2TP + FP + FN